Prior Guidance and Principal Attention Network for Remote Sensing Image Change Detection

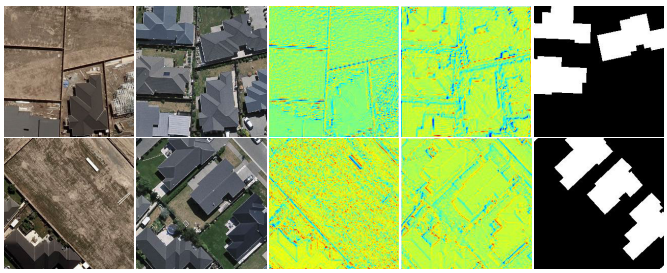
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***Abstract*— In the field of remote sensing (RS) image change detection (CD), the conventional encoder–decoder architecture networks often encounter three significant challenges. First, noise in the features extracted from traditional backbone networks leads to blurred boundaries of change objects. Second, upsampling techniques employed in the decoder, such as interpolation or deconvolution, are limited by their finite receptive fields, making it challenging to accurately distinguish pseudo-changes. Furthermore, how to merge encoder and decoder features with possible semantic gaps for the fine-grained details is a topic worth considering. To address these challenges, we introduce a prior guidance (PG) module that effectively aggregates prior high-level features as a semantic guidance map to guide encoder features for the enhancement of boundary detection. In addition, we design a principal attention (PA) module, which aggregates global information from principal regions through sparse operations and adaptively allocates this information to the upsampled and encoder features. This not only addresses the deficiency of global information in the upsampled features but also reduces the semantic gap between the encoder and decoder by establishing channel dependencies. PA does not divert attention to irrelevant regions, demonstrating excellent performance and computational efficiency. By integrating these two modules into our method, a novel PG and PA network (PGPANet) is elaborately designed. A wide range of experiments confirms the validity of our method, showcasing outstanding detection accuracy on three publicly available CD datasets: LEVIR-CD, SYSU-CD, and WHU-CD. The demo code of this work is publicly available at https://github.com/DaGuangDaGuang/PGPANet.**

***Index Terms*— Change detection (CD), high-level feature aggregation, principal attention (PA), remote sensing (RS), semantic map guidance.**

I. Introduction

CHANGE detection (CD) techniques are employed to discern any alterations within images of the same geographic location captured at various time intervals. As sensor technologies advance swiftly, remote sensing (RS) image data have experienced significant growth, leading to the widespread application of CD techniques in tasks such as vegetation assessment [1], environmental monitoring [2], [3], and disaster assessment [4]. Traditional early CD methods primarily rely on manually crafted features for obtaining change results and require complex classifiers. Over the past few years, the emergence of deep learning techniques has inspired numerous approaches to tackling CD problems. The introduction of Siamese networks, a type of network architecture that compares and extracts features from bi-temporal images, has significantly enhanced feature extraction capabilities, serving as the foundation for many subsequent studies. Several convolutional neural networks (CNNs), such as the UNet series [5] and ResNet series [6], are frequently employed as feature extractors for Siamese networks. Subsequently, these features are progressively restored to high-resolution images through the designed of decoders, resulting in a U-shaped encoder–decoder architecture. However, some primary limitations exist, which continue to impact the detection performance of encoder–decoder networks.



(a) (b) (c) (d) (e)

Fig. 1. Visualization of feature maps extracted from backbone network. (a) T_1 images. (b) T_2 images. (c) and (d) Feature maps extracted from the backbone network. (e) Label.

First, the features from the encoder layers cannot provide sufficient information to achieve accurate boundary detection, they require further refinement [7], [8]. Concretely, due to the conventional employment of regular sliding windows by traditional backbone networks to traverse all local regions of an image, such windows can only capture local contextual information. The diverse and substantial noise within these local representations hinders the identification of semantic boundaries. As indicated by the content within the third and fourth columns of Fig. 1, the noise information in the feature map introduces significant interference to the detection of the change objects. As illustrated in the first row of Fig. 2, obvious errors (highlighted in red) our method demonstrates enhanced performance in such situations.

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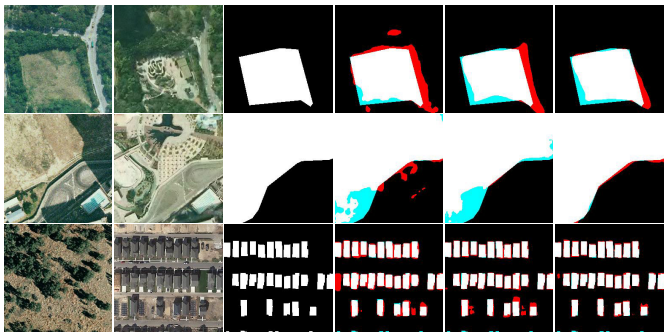


Fig. 2. Challenges in CD tasks. (a) T_1 images. (b) T_2 images. (c) Label. (d) ICIF [14]. (e) DMINet [15]. (f) Ours. In the figure, white represents true positives, red corresponds to false positives, black denotes true negatives, and blue indicates false negatives. These recently proposed methods have failed to accurately detect boundaries, pseudo-changes, and fine-grained details. In contrast, our method reduces the areas of erroneous detection, resulting in improved detection results.

In addition, many commonly employed upsampling methods in decoders, such as bilinear interpolation, deconvolution, and others, only consider limited neighborhoods [2], [9]. This limitation leads to pixel-level blurriness in the feature maps reconstructed from a finite receptive field during the upsampling of high-level feature maps, causing false detection of pseudo-changes. Pseudo-change occurs when unchanged objects are erroneously identified as changed due to differences in spectral and texture information caused by factors such as illumination or viewing angles. The absence of global information within a finite receptive field is highly disadvantageous for pseudo-CD. As depicted in the second row of Fig. 2, variations in spectral conditions result in vegetation appearing in different colors at different times, creating indistinguishable pseudo-changes. Furthermore, some existing decoders typically employ simple addition or concatenation operations to merge encoder and decoder features [10], [11], how to effectively fuse the features with possible semantic gap to detect fine-grained details also worth pondering. Specifically, when performing simple addition or concatenation operations on the feature maps obtained from the encoder and the upsampled outputs of the corresponding decoder network, the feature maps of the decoder at the corresponding size have already undergone multiple convolutions. The existing semantic gap results in the loss of certain crucial information and the interference of irrelevant noise, thereby complicating the detection of fine details. As depicted in the third row of Fig. 2, errors have occurred in detecting fine-grained features within densely built structures.

Concerning the aforementioned primary limitations, some works chose to use heavy backbones, i.e., VGG19 [12] and ResNet50 [6] to enhance feature representation. In [13], channel attention is integrated into the encoder with ResNet as the backbone network to enhance the extraction of semantic features. In [16], atrous separable convolution was introduced to acquire multiscale features. Lei et al. [17] aggregated average pooling at different scales to establish multiscale global features, thereby enhancing edge integrity. Li et al. [18] inserted transformer structures into CNN architectures to capture long-range dependencies. In [19], a pooled-transpose (PT) attention is introduced to capture enhanced global context. Yu et al. [20] combined dilated convolutions with multiple rates (DCMRs) and Conv-Attention to extract multireceptive-field global context information. Many approaches have overlooked the insufficient global context during upsampling operations. Several works integrated transformer structures for the upsampled recovery of feature maps [21], [22], [23], [24], [25]. Overall, for all the mentioned methods, heavy-weight networks like VGG and ResNet50, although capable of extracting richer features that are beneficial for boundary detection, have an extensive parameter count. Although pooling operations such as average pooling and PT attention expand the receptive field, enabling convolutions to capture more information, they also entail information loss during dimensionality reduction, affecting precise boundary detection. Atrous separable convolution or the combination of dilated convolution with attention enlarges the receptive field without pooling-induced information loss but necessitates a significant number of redundant learnable parameters. Utilizing the transformer mechanism as the encoder or embedding attention mechanisms in traditional backbone networks can offer substantial advantages in capturing long-range dependencies and enhancing CD accuracy to varying degrees. However, they come with significant additional memory and computational costs.

To address the aforementioned issues, we propose a framework called prior guidance and principal attention network (PGPANet) in this article. The CD solution is achieved through the following two perspectives.

A. Detecting Accurate Boundaries With a Prior Guidance (PG) Module

Considering a balance between parameter size and hardware consumption, MobileNetV2 [26] is adopted as our backbone network. Specifically, we select the deepest three layers of features extracted from MobileNetV2 as prior information without introducing additional computational structures for attention gating. The prior high-level features aggregation (PHFA) submodule is proposed to integrate the selected features and generate a semantic guidance map, which pinpoint the specific location information of the changed areas. We facilitate interaction among neighboring features and employ the map to guide feature interactions for pinpointing specific areas of change, helping the features to acquire precise boundary information. Consequently, we exclusively input refined encoder features into the decoder, thereby avoiding the incorporation of redundant noise.

B. Designing a Lightweight Decoder With a Principal Attention (PA) Module

First, we design a global-level information dynamic pruning (GIDP) submodule to aggregate global information. In practice, we use GIDP to select the top-k principal spatial regions from the lowest resolution feature map and at the same time, choose the top-k principal channels from the features obtained through simple upsampling. By aggregating the most

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principal information in both spatial and channel, we utilize the top-k operation to trim away nonrelevant information from the global context. In contrast to the high computational cost (H^2W^2) associated with traditional self-attention mechanisms, our GIDP mechanism offers computational efficiency, with a complexity of $O(CkHW)$, noting that $k << HW$. Then, sparse operations are employed to prevent the dispersion of attention to unrelated regions. The sparse-level attention adaptive allocation (SAAA) submodule is crucial, as it adaptively distributes global information to upsampled features, compensating for the insufficient global cues during the upsampling process. In addition, we allocate sparse-level attention to construct the channel dependencies between encoder and decoder features, reduce the semantic gap, and better detect fine-grained details.

Our main contributions are summarized as follows.

1) A PG module is introduced to fully leverage the characteristics of high and low-level features, addressing the issue of missing precise boundaries.
2) A lightweight PA module is designed to tackle the issue of false pseudo-CD caused by traditional simple upsampling methods in the decoder. It also bridges the semantic gap inherent in the encoder–decoder structure, thereby enhancing the detection of fine-grained detail information.
3) Extensive experiments conducted on the proposed PGPANet demonstrate its superior performance compared to thirteen other state-of-the-art methods, with relatively low parameter and computational requirements, rendering it suitable for resource-constrained environments.

II. Related Work

A. Traditional CD Methods

In early CD algorithms, algebraic-based methods predominantly employed strategies like image regression [27], [28], and image differencing [29] to obtain difference maps. In image change-based methods, techniques such as change vector analysis (CVA) [30], principal component analysis (PCA) [31], and tasseled cap transformation (KT) [32] were typically used to obtain intrinsic features of changed objects. Due to the lower resolution and acquisition challenges associated with early RS images, clustering-based algorithms also found widespread application. These algorithms clustered images into two categories: changed and unchanged, to accomplish CD tasks. Classical clustering methods include fuzzy c-means clustering (FCM) [33], Markov random field (MRF) fuzzy clustering [34], and more. However, these traditional methods suffered from high manual dependence and poor robustness, making them less effective in addressing CD tasks in complex scenarios.

B. Deep-Learning CD Methods

With the development of deep learning, U-shaped architectures [5] with encoder–decoder structures have found widespread application in current CD methods. Three different U-net-based models, namely, FC-Siam-conc [35], FC-Siam-diff [35], and FC-early fusion (EF) [35], were proposed for obtaining bi-temporal difference maps in this context. Regarding the design of encoder–decoder structures, researchers have explored various strategies. Various attention mechanisms, including spatial attention [36], channel attention [37], and cross attention [38], [39] were introduced to better extract positional information of changes. Fan et al. [13] employed a channel attention mechanism to improve semantic feature representation in the encoder stages and utilized deformable convolutions to detect change boundaries. Fang et al. [40] introduced ensemble CA for deep supervision. In approach [41], the introduction of atrous convolutions, allowed convolutional kernels to skip a certain number of pixels while processing input, thus covering a broader spatial range and capturing multiscale features. A2Net [42] designed an aggregation strategy for adjacent features, effectively compensating for the shortcomings in feature extraction by the encoder. TFI-GR [43] adopted a strategy of global average pooling by averaging features of different scales, thus improving the aggregation of temporal difference information. Lei et al. [44] designed a multiscale decoupled convolution that decouples a regular convolution into the concatenation of spatial correlation and channel correlation. The channel correlation is computed through pointwise convolution, while the spatial correlation learns feature information by iteratively using combinations of dilation rates in depthwise convolution. SNUNet [40] integrated the concepts of UNet++ [45] into CD, introducing nested and dense skip connections in the encoder and decoder to reduce the semantic gap between them. However, it still failed to explore sufficient information from multiple scales. Miao et al. [46] proposed a full-scale connected UNet-based network, SNUNet3+, which employs full-scale skip connections to integrate all fine-grained spatial information and coarse-grained semantic information. This approach addresses the issue of insufficient information utilization during feature extraction in the encoder.

Recognizing the outstanding performance of transformers in handling long-range dependencies, some studies, such as BIT [21], have utilized transformer structures to capture distant relationships between features extracted by CNNs. ChangeFormer [22], on the other hand, employed transformer structures to capture bi-temporal feature maps and decode them using a simple MLP structure. TransUNetCD [18] combined CNN and transformer structures in a hybrid network to further reduce false positives and false negatives. SwinSUnet [23] adopted a pure transformer structure for both encoding and decoding, enabling the model to establish a global contextual understanding and handle information associations across distant locations. ICIF [14] designed a network that integrates CNNs and transformers for feature extraction, thereby capturing both local and global features. DMINet [15] integrated two distinct attention mechanisms to capture global features while suppressing irrelevant task interference. Zhang et al. [25] constructed two cross-temporal difference transformer encoders to extract features of change areas in a cross manner from the embedded tokens of multitemporal images, capturing relational changes in the scene.

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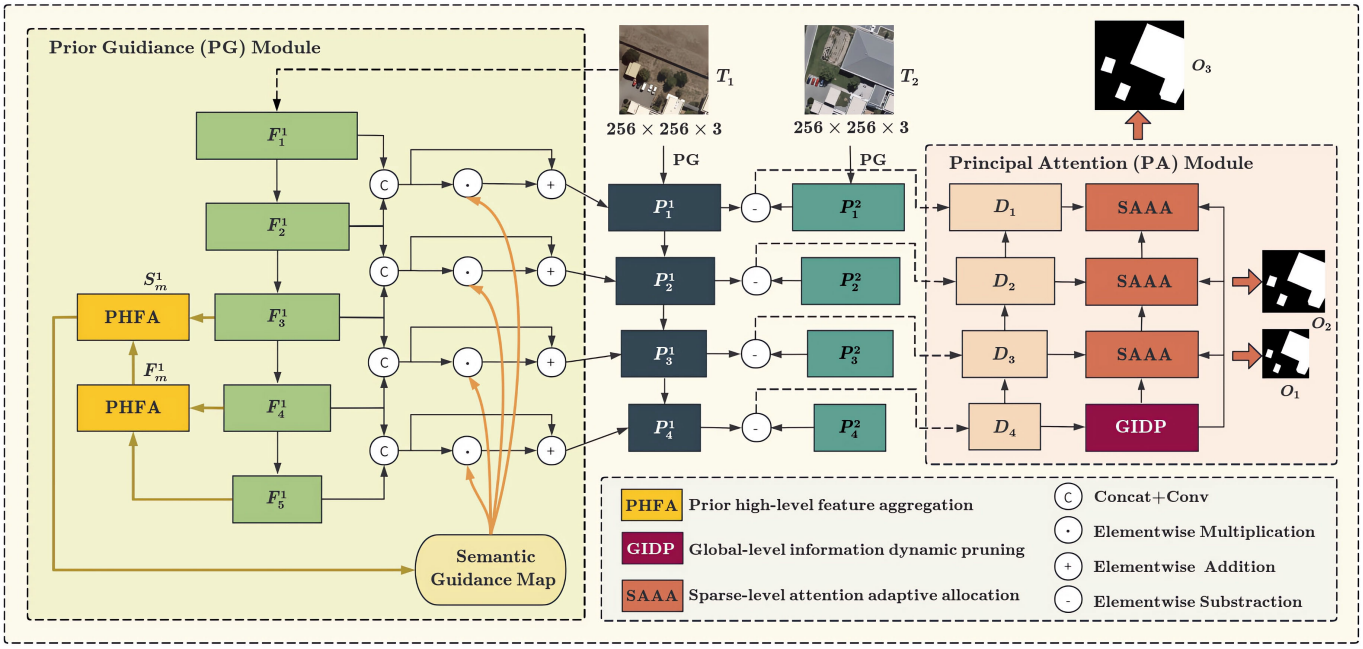


Fig. 3. Framework of the proposed PGPANet. The temporal features are obtained from the pretrained MobileNetV2 [26]. Then, we use the PG module to reduce redundant noise information in the encoder and locate the areas of change, achieving precise change boundaries for CD. PA is designed to tackle the problem of false CD caused by traditional simple upsampling methods and remove the semantic gap between the encoder and decoder. Finally, change maps are obtained after the PA module.

Shangguan et al. [47] proposed a branch transformer block (BTB), which designs two input branches to merge semantic labels and spatial features. The multihead self-attention in the branches of BTB can effectively enhance self-attention learning, making it suitable for large-scale inputs. Pan et al. [48] introduced a novel Siamese transformer encoder that effectively detects changes in small-scale buildings by integrating multiscale features obtained from different windows through hierarchical window fusion.

As we know low-level features encompass accurate boundary details related to changing objects, while high-level features contain positional information [10], [49], [50]. However, in the encoder part, existing works have not fully exploited the differences between high-level and low-level features, or they have adopted complex structures to mitigate the issue of insufficient information utilization during feature extraction. Our PG module addresses this by aggregating prior high-level features as a semantic guidance map to guide the encoder features, thereby effectively acquiring precise boundary information and tightly integrated internal information. Regarding the decoder structure, although techniques such as dilated convolutions, global average pooling, and transformer structures have enhanced the ability to capture global information, they have also introduced considerable hardware requirements. Furthermore, these methods overlooked addressing the issue of pseudo-CD from the perspective of enhancing global information during the upsampling process and did not fully consider capturing fine-grained details within the regions of change. In contrast, our PA module adaptively allocates global information to upsampled features and establishes dependencies between the encoder and decoder. Based on the above discussion, we designed a novel network named PGPANet.

III. Method

The overall framework of our PGPANet is illustrated in Fig. 3, where the network receives the bi-temporal images as input and comprises two essential components: PG module and PA module.

Encoder: Given the bi-temporal images T_1 and T_2, the five layers of features extracted can be denoted as $F_i^t \in \mathbb{R}^{C_i \times jH \times jW} (i \in 1, 2, 3, 4, 5, j \in 2^{4-i}, t \in 1, 2)$. We propose a PG module to enhance feature representation for the detection of accurate boundaries. The specifics of the PG module are provided in Section III-A and this procedure can be formulated as follows:

$$P_i^t = \Phi(F_1^t, F_2^t, F_3^t, F_4^t, F_5^t), \quad t \in 1, 2, i \in 1, 2, 3, 4 \tag{1}$$

where $\Phi(\cdot)$ denotes the function of PG, and $P_i^t \in \mathbb{R}^{C_i \times jH \times jW} (i \in 1, 2, 3, 4, j \in 2^{4-i})$ represents the enhanced bi-temporal features. Subsequently, utilizing the feature difference method, we perform an absolute elementwise subtraction operation on the bi-temporal features to generate difference features $D_i \in \mathbb{R}^{C_i \times jH \times jW} (i \in 1, 2, 3, 4, j \in 2^{4-i})$

$$D_i = \left| P_i^1 \ominus P_i^2 \right|, \quad i \in 1, 2, 3, 4 \tag{2}$$

where $|\cdot|$ denotes the absolution operation and $\ominus$ represents the elementwise subtraction operation.

Decoder: We propose a PA module to address the challenges of identifying pseudo-changes within changed objects under different spectra and reducing the disparity in semantics between the encoder and decoder features. The PA module architecture will be elaborated in Section III-B. Our decoder formulation is as follows:

$$O_i = \Psi(D_1, D_2, D_3, D_4), \quad i \in 1, 2, 3 \tag{3}$$

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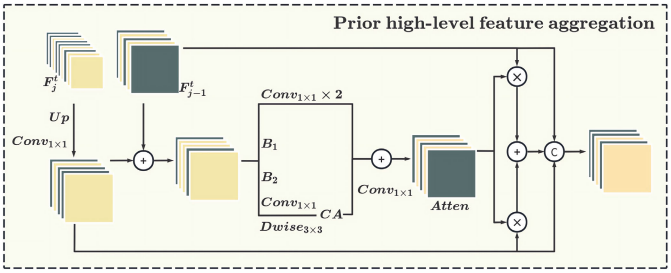


Fig. 4. Illustration of the proposed PHFA submodule. PHFA is used to integrate the selected features and generate a semantic guidance map, which pinpoint the specific location information of the changed areas.

where $\Psi(\cdot)$ denotes the PA module and $O_i \in \mathbb{R}^{C_i\times jH\times jW}(i \in 1, 2, 3, j \in 2^{4-i})$ is the outputs.

A. PG Module

In consideration of the distinct characteristics of high-level and low-level features, we opt to aggregate high-level features as prior information to generate a semantic guidance map for change region localization. This map is utilized to provide positional information about the detected change regions, guiding the extraction of boundaries from low-level features, thereby enriching the internal information within the change regions and enhancing their cohesiveness. This module primarily consists of two steps, and the details are as follows.

1) Prior High-Level Features Aggregation: As illustrated in Figs. 3 and 4, here, we take the features extracted from the backbone network using the T_1 image as an example. Compared to directly feeding features into the decoder through skip connections, we commence by selecting the deepest three layers with enriched semantic information as our prior knowledge. Specifically, we upsample the feature map $F_5^1 \in \mathbb{R}^{C_5\times H/2\times W/2}$ to match the size of $F_4^1 \in \mathbb{R}^{C_4\times H\times W}$ and then reduce the number of channels using a 1×1 convolutional layer. The adjusted F_5^1 is then elementwise added to F_4^1, resulting in the amalgamated initial feature $F_{45}^1 \in \mathbb{R}^{C_4\times H\times W}$, formulated as

$$\widetilde{F_5^1} = \mathrm{Conv}_{1\times1}(\mathrm{Upsampling}(F_5^1)) \tag{4}$$

$$F_{45}^1 = F_4^1 \oplus \widetilde{F_5^1} \tag{5}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ denotes to a 1×1 convolutional layer followed by batch normalization and a ReLU activation, Upsampling$(\cdot)$ denotes a bilinear upsampling operation, and $\oplus$ is the elementwise addition operation.

Then, the fused initial features undergo a dual-branch operation. We use a dual-layer convolution in the first branch to extract textural characteristics from the initial features, thereby retaining semantic details from high-level features. For the second branch, channel expansion is performed through the application of a 1×1 convolutional layer, followed by a 3×3 depthwise separable convolution to further extract features. It is noteworthy that while traditional SE [51] attends to inter-channel information interactions, it overlooks the significance of spatial positional information. Although methods such as BAM [52] and CBAM [53] endeavor to introduce positional information through global pooling along channel dimensions, these approaches are limited to capture local information, failing to capture spatial dependencies over distant regions. Therefore, we adopt a coordinate attention mechanism [54] to embed positional information. By introducing this mechanism along the vertical and horizontal spatial directions, we are able to focus on crucial spatially changed regions. Ultimately, we employ a 1×1 convolution to affect channel reduction. The process is given as follows:

$$B_1 = \mathrm{Conv}_{1\times1}(\mathrm{Conv}_{1\times1}(F_{45}^1)) \tag{6}$$

$$B_2 = \mathrm{Conv}_{1\times1}(\mathrm{CA}(\mathrm{Dwise}_{3\times3}(\mathrm{Conv}_{1\times1}(F_{45}^1)))) \tag{7}$$

where $B_1 \in \mathbb{R}^{C_4\times H\times W}$ and $B_2 \in \mathbb{R}^{C_4\times H\times W}$ represent the first and second branches, respectively, $\mathrm{Dwise}_{3\times3}(\cdot)$ denotes a 3×3 depthwise separable convolutional layer equipped with batch normalization and a ReLU activation function, and $\mathrm{CA}(\cdot)$ represents the coordinate attention mechanism [54].

To this end, the information from the dual branches is initially aggregated through an elementwise addition operation, followed by a sigmoid activation to generate the attention map for the dual-layer features. The attention map is then elementwise multiplied with the feature maps F_4^1 and $\widetilde{F_5^1}$, producing an enhanced feature representation $\widetilde{F_{45}^1} \in \mathbb{R}^{C_4\times H\times W}$. The specific operations are as follows:

$$\mathrm{Atten} = \mathrm{sigmoid}(B_1 \oplus B_2) \tag{8}$$

$$\widetilde{F_{45}^1} = \mathrm{Atten} \odot F_4^1 \oplus (1 \ominus \mathrm{Atten}) \odot \widetilde{F_5^1} \tag{9}$$

where $\mathrm{Atten} \in \mathbb{R}^{C_4\times H\times W}$ represents the attention map and $\odot$ denotes the elementwise multiplication operation. Then, the $\widetilde{F_{45}^1}$ is combined with the original features by a feature concatenation process, facilitating the further aggregation of information across the two hierarchical levels as follows:

$$F_m^1 = \mathrm{Conv}_{1\times1}(\mathrm{Cat}(F_4^1, \widetilde{F_{45}^1}, \widetilde{F_5^1})), \tag{10}$$

where $\mathrm{Cat}(\cdot)$ is the concatenation operation. The same procedure can be applied to $F_3^1 \in \mathbb{R}^{C_3\times2H\times2W}$ and $F_m^1 \in \mathbb{R}^{C_4\times H\times W}$ using the PHFA strategy, yielding the ultimate semantic guidance map S_m^1.

2) Semantic Map Guiding: As shown in the left of Fig. 3, we interact with the neighboring features through the concatenation operation to obtain $\widetilde{F_i^1} \in \mathbb{R}^{C_i\times jH\times jW}(i \in 1, 2, 3, 4, j \in 2^{4-i})$, and then multiply the semantic guidance map S_m^1 with the interacted features, respectively. Concurrently, we adopt a residual connection branch, further amplifying the prowess of feature representation. The process is as follows:

$$\widetilde{F_i^1} = \mathrm{Conv}_{1\times1}(\mathrm{Cat}(F_i^1, F_{i+1}^1)), \quad i \in 1, 2, 3, 4 \tag{11}$$

$$P_i^1 = \widetilde{F_i^1} \odot S_m^1 \oplus \widetilde{F_i^1}, \quad i \in 1, 2, 3, 4. \tag{12}$$

Thus, the PG mechanism for the posttemporal image can be calculated in the same way, resulting in the enhanced temporal features $P_i^2(i \in 1, 2, 3, 4)$.

B. PA Module

Our key idea is to dynamically prune most of the irrelevant information at a global-level, thereby selecting information from a small but principal subset of parts, which utilizes sparse operation to save both computation and memory. The

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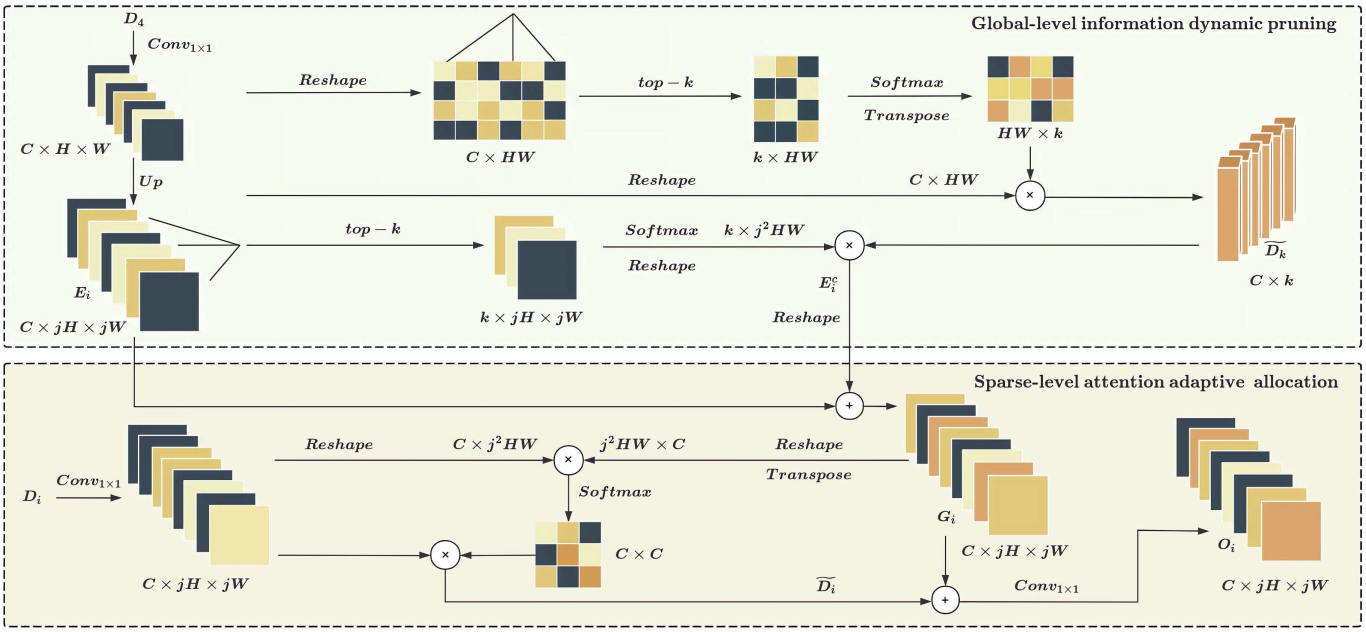


Fig. 5. Illustration of the presented PA module. PA is designed to address the issue of false pseudo-CD induced by traditional simple upsampling techniques within the decoder. Furthermore, it endeavors to bridge the semantic gap that exists in the encoder–decoder structure, thereby facilitating the acquisition of more refined and detailed information.

Algorithm 1 Pseudo-Code of GIDP in a PyTorch-Like Style

Input: D_4 (C, H, W). Assume $H == W$
Output: E_i^c (C, j^2HW), $i \in 1, 2, 3$, $j = 2^{4-i}$

compute matrix of principal regions
k: numbers of the k values, B: batch size
bmm: batch matrix multiplication
$\underline{A_k}$: attention map, (k, HW)
$\widetilde{D_k}$: sparse-level attention map, (C, k)
$\boldsymbol{A_k}$ = topk(input.view(B, C, -1), k, dim = 1)[0]
$\boldsymbol{A_k}$ = softmax(A_k, dim = -1)
$\boldsymbol{D_r}$ = input.view(B, C, -1)
$\widetilde{\boldsymbol{D_k}}$ = bmm(D_r, A_k.permute(0, 2, 1))

compute matrix of principal channels
E_i: features upsampled from D_4, (C, jH, jW), $i \in 1, 2, 3$, $j = 2^{4-i}$
E_i^k: sparse-level attention, (C, jH, jW), $i \in 1, 2, 3$, $j = 2^{4-i}$
$\boldsymbol{E_i^k}$ = topk(E_i, k, dim = 1)[0]
$\boldsymbol{E_i^k}$ = softmax(E_i^k, dim = -1)

fused sparse-level attention representation
$\boldsymbol{E_i^c}$ = bmm($\widetilde{D_k}$, E_i^k.view(B, k, -1)).view(B, -1, H, W)

derived sparse-level attention can be utilized to offset the deficiency of global information within the upsampled features and establish channel dependencies between the encoder and decoder. A detailed explanation is as follows.

1) Global-Level Information Dynamic Pruning: The algorithm of the GIDP submodule is summarized in Algorithm 1. As shown in the top box of Fig. 5, given that the lowest resolution $D_4 \in \mathbb{R}^{C_4 \times H \times W}$ as the first input to PA module, a 1×1 convolutional layer is utilized to consolidate its channel into $C(C = 64$, in our approach), and then a reshape operation is performed to obtain the matrix $D_r \in \mathbb{R}^{C \times HW}$. Technically, for the matrix D_r, we retain only the top-k (e.g., 1, 16, 32, 64) principal spatial regions. A top-k operator is employed to get the pruned region-level principal information $D_k \in \mathbb{R}^{k \times HW}$ by pruning the irrelevant information and preventing the dispersion of attention to unrelated regions. Then, a softmax operation is used to obtain the attention map $\underline{A_k} \in \mathbb{R}^{k \times HW}$. Finally, the preliminarily sparse-level attention $\widetilde{D_k} \in \mathbb{R}^{C \times k}$ is obtained through matrix multiplication between D_r and transposed A_k. The operations are as follows:

$$A_k = \text{softmax}(\text{topkIndex}(\text{Re}(\text{Conv}_{1\times1}(D_4)))) \tag{13}$$

$$\widetilde{D_k} = D_r \otimes (A_k)^{\text{T}} \tag{14}$$

where topkIndex$(\cdot)$ represents a top-k operator, Re$(\cdot)$ denotes the reshape operation, and $\otimes$ is the matrix multiplication operation. $E_i \in \mathbb{R}^{C \times jH \times jW}(i \in 1, 2, 3, j = 2^{4-i})$ denotes the feature map progressively upsampled from D_4. For each upsampled feature E_i, we select the top-k principal channels and transform them into an attention matrix $E_i^k \in \mathbb{R}^{k \times j^2HW}$ after a softmax operation. Subsequently, a matrix multiplication operation is used between the obtained attention matrix E_i^k and $\widetilde{D_k}$ and obtained the fused sparse-level attention representation $E_i^c \in \mathbb{R}^{C \times j^2HW}$, which contains the most principal information in both spatial regions and channels

$$E_i^k = \text{softmax}(\text{Re}(\text{topkIndex}(E_i))) \tag{15}$$

$$E_i^c = \widetilde{D_k} \otimes E_i^k. \tag{16}$$

Our GIDP seeks a dynamically-aware sparse attention mechanism. Both A_k and E_i^k focus on a small subset of the most semantically relevant elements. For instance, if the attention

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map exhibits high activation values for foreground objects (such as buildings, roads, and vegetation), it will capture more representations from these corresponding foreground objects.

2) Sparse-Level Attention Adaptive Allocation: As illustrated in the bottom box of Fig. 5, the reshaped E_i^c is added to E_i, adaptively allocating sparse-level attention to each level of the upsamped features across the entire spatial domain, generated the transparent pixel-level feature representation $G_i \in \mathbb{R}^{C\times jH\times jW}(i \in 1,2,3,\ j = 2^{4-i})$ for the detection of pseudo-changes. The process is as follows:

$$G_i = \mathrm{Re}(E_i^c) \oplus E_i. \tag{17}$$

Finally, we unify the channel of encoder outputs $D_i \in \mathbb{R}^{C_i\times jH\times jW}(i \in 1,2,3,\ j = 2^{4-i})$ into C, we then establish channel dependencies between G_i and the encoder outputs, allocating sparse-level attention to encoder features, eliminating the semantic gap to retain portions that helps to identify the fine-grained details. This allows us to obtain relevant channels with strong dependencies between the encoder and decoder, resulting in encoder-enhanced features $\widetilde{D_i} \in \mathbb{R}^{C_i\times jH\times jW}(i \in 1,2,3,\ j = 2^{4-i})$. In addition, we fuse G_i with $\widetilde{D_i}$ using an elementwise addition, allowing it to capture pixel-level feature representations and enhance the encoder features for more precise change detail depiction. The output is subsequently fed into a 1×1 convolutional layer, generating the output $O_i \in \mathbb{R}^{C\times jH\times jW}(i \in 1,2,3,\ j = 2^{4-i})$ of the PA module as follows:

$$\widetilde{D_i} = \mathrm{softmax}(\mathrm{Re}(\mathrm{Conv}_{1\times1}(D_i)) \otimes (\mathrm{Re}(G_i))^{\mathrm{T}}) \otimes D_i \tag{18}$$

$$O_i = \mathrm{Conv}_{1\times1}(G_i \oplus \widetilde{D_i}). \tag{19}$$

C. Loss Function

As is known that in CD tasks, the majority of cases involve no change, with the changed objects comprising only a small fraction. Therefore, we have opted for utilizing a mixture of binary cross-entropy loss, denoted as L_{bce}, and the dice loss, denoted as L_{dice} [10], as the loss functions in our model. The total loss function L_{total} is computed as follows:

$$L_{\mathrm{bce}}(O,G) = G \cdot \log O + (1-G)\cdot \log(1-O) \tag{20}$$

$$L_{\mathrm{dice}}(O,G) = 1 - \frac{2\cdot O\cdot G}{\|O\|_1 + \|G\|_1} \tag{21}$$

$$L_{\mathrm{total}} = \sum_{i=1}^{3}(L_{\mathrm{bce}}(O_i,G) + L_{\mathrm{dice}}(O_i,G)) \tag{22}$$

where O_i represents the predicted output, G is the corresponding ground truth, and $\|\cdot\|_1$ denotes the l_1 norm.

IV. Experiments

A. Experimental Setup

1) Implementation Details: Our PGPANet is implemented in the Pytorch framework and conducted experiments on a single GeForce RTX 1080Ti GPU with 12-GB VRAM. We employ a MobileNetV2 [26], pretrained on ImageNet, as our backbone network for extracting hierarchical temporal features. During training, we employ Adam with a momentum of 0.9 for model optimization [58], and the maximum number of iterations is fixed at 160 000. The starting learning rate is initialized at 0.0001, and the batch size is set to 8.

2) Evaluation Metrics: We have selected four widely used metrics to assess the efficacy of our experiments, including recall (Rec), precision (Pre), $F1$-score ($F1$), and intersection over union (IoU). The following are the specific formulas for defining the evaluation metrics as mentioned above:

$$\mathrm{Rec} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \tag{23}$$

$$\mathrm{Pre} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \tag{24}$$

$$F1 = 2/(\mathrm{Rec}^{-1} + \mathrm{Pre}^{-1}) \tag{25}$$

$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}) \tag{26}$$

where TP, TN, FP, and FN denote the counts of true positives, true negatives, false positives, and false negatives, respectively.

B. Datasets

To ascertain the efficacy of our proposed PGPANet, we performed a comprehensive range of experiments on the following three datasets, with details of each dataset described as follows.

1) LEVIR-CD [55]: The dataset is a publicly available CD dataset, consisting of 637 pairs of RS images, each at a resolution of 1024 × 1024 pixels. We process these images by splitting them into nonoverlapping patches of 256 × 256 in size, which yields 7120 patches for training, 1024 for validation, and 2048 for testing.

2) SYSU-CD [56]: This dataset offers a diverse collection of intricate change scenarios, which comprises 20 000 pairs of bi-temporal RS image patches, each with a spatial size of 256 × 256 pixels. The SYSU dataset follows an official split ratio of 6:2:2, which yields 12 000 patches for training, 4000 for validation, and 4000 for testing.

3) WHU-CD [57]: This dataset mainly targets regions that have been reconstructed after experiencing earthquakes. We divide these images into nonoverlapping patches of 256 × 256 pixels and allocate them into three separate groups for training, validation, and testing, which gives us 6096, 762, and 762 samples for each subset, respectively.

C. Comparison With State-of-the-Art Methods

1) Quantitative Evaluation: We compared our proposed method with nine state-of-the-art CD techniques. To ensure a fair comparison, we replicated these methods using their publicly available code with default parameters. Table I presents a quantitative evaluation of various methods on three CD datasets, including IoU, $F1$, Rec, and Pre. Furthermore, we detailed the model parameters (Params) and computational costs (FLOPs) in Table I.

Results presented in Table I indicate the superior efficacy of our proposed PGPANet in comparison to other networks. For instance, on the WHU-CD dataset, our PGPANet achieved a 1.41% and 0.79% improvement in IoU and $F1$, respectively. Persistent enhancements are noticeable across various datasets and evaluation metrics. Notably, our proposed method achieved better results with relatively low parameter and computational requirements. For example, on the

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TABLE I

COMPARISON OF QUANTITATIVE PERFORMANCE OF VARIOUS CD TECHNIQUES. ALL METRICS ARE EXPRESSED AS PERCENTAGES (%) WITH OUTSTANDING PERFORMANCES MARKED IN **BOLD**

Model	Params (M)	FLOPs (G)	LEVIR-CD [55]				SYSU-CD [56]				WHU-CD [57]			
			IoU	F1	Rec	Pre	IoU	F1	Rec	Pre	IoU	F1	Rec	Pre
FC-diff [35]	1.35	9.42	75.18	85.83	84.59	87.12	42.05	59.20	46.72	80.80	69.48	81.99	78.90	89.14
FC-ef [35]	1.35	7.13	71.80	83.59	82.27	84.94	55.83	71.65	65.41	79.22	58.53	71.65	65.41	79.22
FC-cat [35]	10.32	1.55	76.98	86.99	85.64	88.38	60.68	75.53	71.38	80.21	72.24	83.87	84.22	83.53
SNUNet [40]	8.45	34.63	82.20	90.23	88.97	91.54	66.73	80.04	79.79	80.30	76.02	86.38	88.95	83.95
BIT [21]	3.01	16.88	82.06	90.15	88.98	91.34	62.50	76.93	72.50	81.93	80.40	89.13	89.69	88.58
ChangeFormer [22]	41.03	202.79	81.93	90.07	87.38	**92.92**	60.46	75.36	71.06	80.22	83.88	91.23	90.76	91.71
TFI-GR [43]	28.37	9.73	83.03	90.73	89.53	91.95	71.38	83.30	81.16	85.56	88.55	93.92	92.65	95.24
ICIF [14]	23.82	25.36	79.86	88.80	86.85	90.84	61.38	76.07	73.31	**91.94**	78.27	87.81	84.06	91.96
DMINet [15]	14.42	6.42	83.14	90.79	88.89	92.78	68.20	81.10	76.82	85.88	79.53	88.60	91.24	86.10
USSFC-Net [44]	1.52	4.86	82.54	90.44	91.82	89.09	65.75	79.34	81.52	77.27	83.92	91.26	93.20	89.40
M-Swin [48]	-	-	83.57	91.05	90.05	92.07	-	-	-	-	86.09	92.94	91.37	94.56
ELGC-Net [19]	10.57	123.59	83.52	91.02	89.78	92.30	65.34	79.03	77.26	80.89	75.56	86.08	82.15	90.40
GCFormer [20]	35.43	11.30	83.21	90.83	**92.50**	89.34	-	-	-	-	83.43	90.97	90.15	91.81
Ours	4.13	13.86	**84.54**	**91.62**	91.52	91.72	**72.23**	**83.87**	**83.33**	84.43	**89.96**	**94.71**	**93.91**	**95.53**

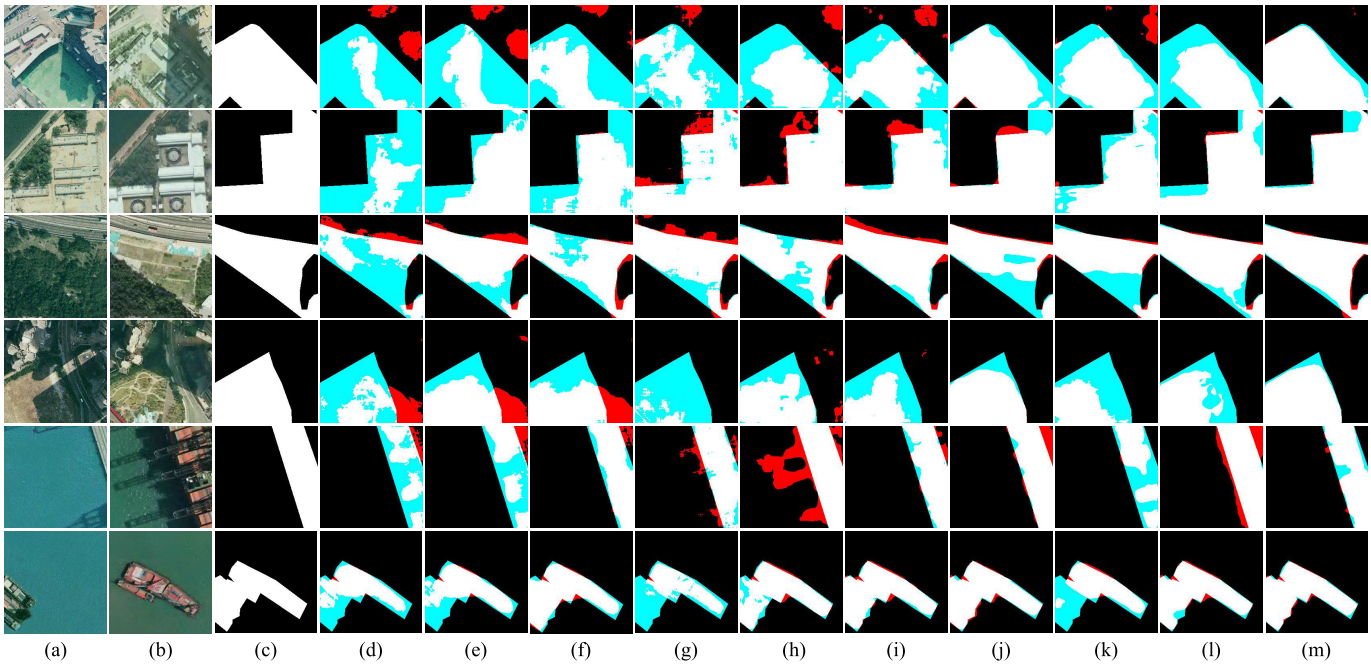


Fig. 6. Qualitative visualizations on SYSU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Label. (d) FC-diff. (e) FC-ef. (f) FC-cat. (g) SNUNet. (h) BIT. (i) ChangeFormer. (j) TFI-GR. (k) ICIF. (l) DMINet. (m) Ours.

LEVIR-CD dataset, the ICIF method required 23.82M parameters and 25.36G FLOPs to achieve an $F1$-score of 88.80, while our method achieved an $F1$-score of 91.62 with only 4.13M parameters and 13.86G FLOPs. Despite ChangeFormer, SNUNet, and ELGC-Net obtained relatively good scores, their model is quite large. While some other methods are relatively lightweight(FC-diff, FC-ef, FC-cat, and USSFC-Net), their accuracy still needs improvement and they may not perform well in complex scenarios. In contrast, our approach achieved the best results and reduced the parameter and computational requirements.

2) Qualitative Evaluation: Experimental results of various approaches have been subject to visual analysis through Figs. 6–8. For improved visual distinctiveness, we employed different colors to denote various detection outcomes: true positives are shown in white, false positives in red, true negatives in black, and false negatives in blue. Based on the experimental findings, we observe outstanding performance in multiple aspects of the method we proposed.

1) Our method effectively distinguishes pseudo-changes. As shown in the first, fourth, and fifth rows of Fig. 6, interference from shadows formed under sunlight results in the same buildings exhibiting varying colors at different times, causing the change areas to become blurred and leading to irrelevant changes. Many methods are susceptible to these disturbances, resulting in erroneous detection results. In contrast, our method adeptly handles pseudo-changes, accurately detecting genuine change areas. In the second and third rows of Fig. 6, newly appearing changes bear similarity to road or vegetation

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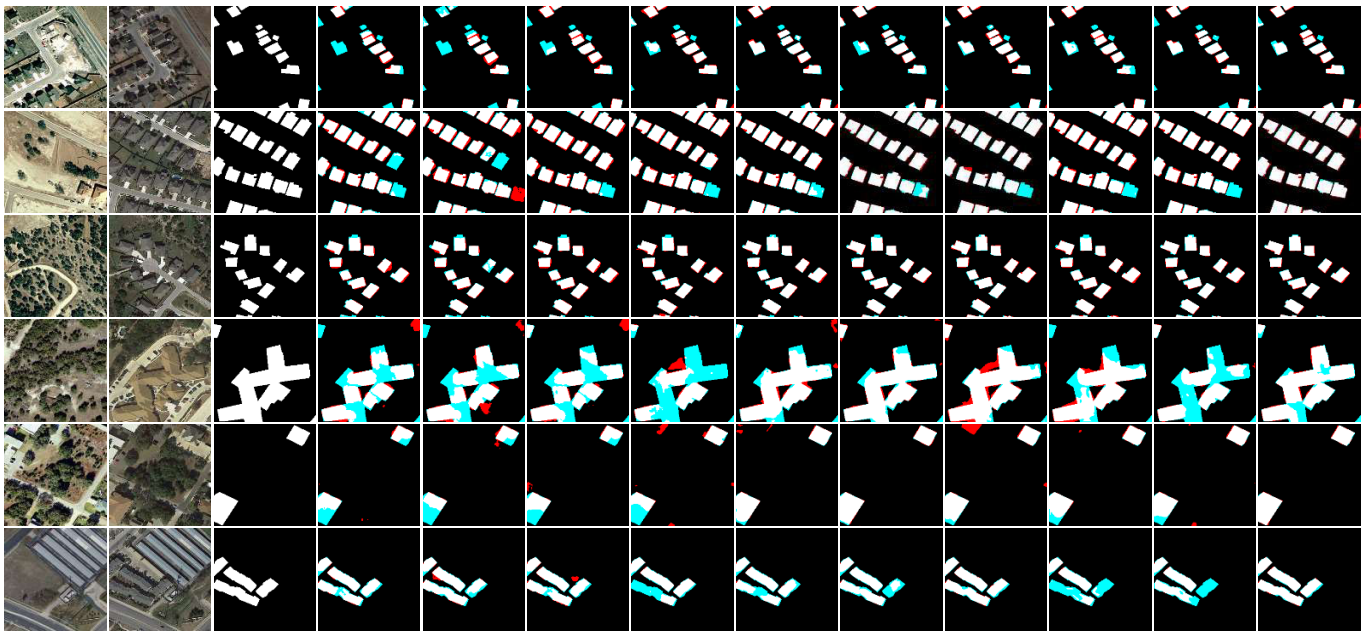


Fig. 7. Qualitative visualizations on LEVIR-CD dataset. (a) T_1 images. (b) T_2 images. (c) Label. (d) FC-diff. (e) FC-ef. (f) FC-cat. (g) SNUNet. (h) BIT. (i) ChangeFormer. (j) TFI-GR. (k) ICIF. (l) DMINet. (m) Ours.

colors, while in the sixth row of Fig. 7, the colors of new buildings closely resemble ground surface colors. Many methods struggle to differentiate such changes, but our approach effectively addresses pseudo-changes arising from the diversity of land cover appearances.

2) Our method excels at preserving fine-grained structural details. As shown in Fig. 7, precise detection of small-scale structures is crucial, particularly when dealing with rugged road changes, as evident in the first, second, third, and fourth rows of Fig. 7. In such cases, FC-diff, FC-ef, FC-cat, BIT, and DMINet exhibit notable omissions, and ICIF encounters issues in the detection of red regions. In contrast, our method comprehensively identifies detailed regions, achieving results closely aligned with ground truth labels. As demonstrated in the fifth and sixth rows of Fig. 7, many methods exhibit errors in detecting subtle structures, whereas our approach performs well.

3) Our method exhibits a significant advantage in precise boundary recognition. As illustrated in Fig. 8, in comparison to other high-performance methods, such as ChangeFormer, TFI-GR, ICIF, and DMINet, our method excels in suppressing background interference in bitemporal images and demonstrates heightened sensitivity in detecting boundaries, leading to a substantial reduction in boundary omissions and erroneous detection.

D. Ablation Studies

To confirm the impact of various components within our proposed approach, we conducted a set of detailed ablation studies across three CD datasets in this section. The experiment results are shown as follows.

TABLE II

ABLATION STUDIES OF PG AND PA MODULES WHEN TRAINED ON THREE CD DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	PG	PA	LEVIR-CD	SYSU-CD	WHU-CD
(a)	×	×	88.62	81.65	91.04
(b)	×	✓	90.64	82.69	93.76
(c)	✓	×	90.46	82.32	93.27
(d)	✓	✓	**91.62**	**83.87**	**94.71**

TABLE III

EFFECTIVENESS OF PG MODULE WHEN TRAINED ON THREE CD DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	Aggregation Levels			LEVIR	SYSU	WHU
	F3	F4	F5	-CD	-CD	-CD
-	×	×	×	90.64	82.69	93.76
Add&Concat	✓	✓	×	90.89	82.97	94.04
Add&Concat	×	✓	✓	91.17	83.26	94.21
Add&Concat	✓	✓	✓	91.44	83.58	94.4
PHFA	✓	✓	×	91.27	83.33	94.32
PHFA	×	✓	✓	91.54	83.6	94.57
PHFA	✓	✓	✓	**91.62**	**83.87**	**94.71**

1) Effectiveness of PG Module: As illustrated in Table II, initially, we removed the embedding of the PG module and employed MobileNetV2 [26] for feature extraction, resulting in the lowest scores. When only the PG module is added, the $F1$-score increased by 1.84%, 0.67%, and 2.23% on the LEVIR-CD, SYSU-CD, and WHU-CD datasets, respectively. As shown in Table III, different combinations of high-level features were selected for aggregation. As evident from the fourth and seventh rows in the table, optimal performance was achieved when the deepest three layers of features were chosen to generate semantic guidance maps. To further

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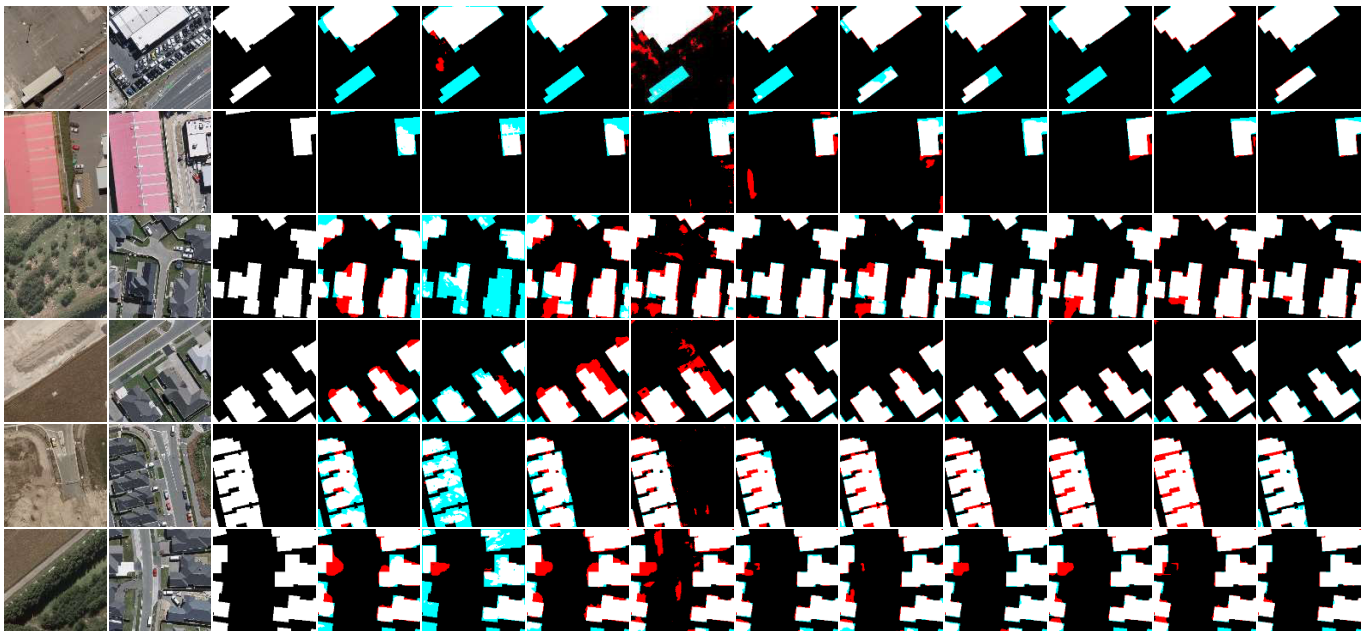


Fig. 8. Qualitative visualizations on WHU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Label. (d) FC-diff. (e) FC-ef. (f) FC-cat. (g) SNUNet. (h) BIT. (i) ChangeFormer. (j) TFI-GR. (k) ICIF. (l) DMINet. (m) Ours.

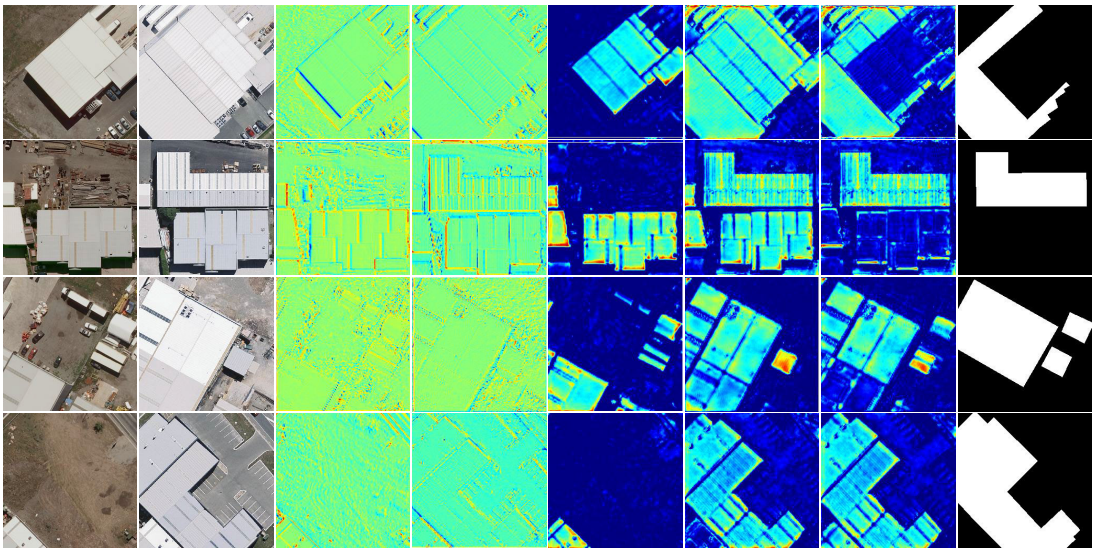


Fig. 9. Visualization of attention maps of PG module. (a) T_1 image. (b) T_2 image. (c) and (d) Feature maps extracted from the backbone network. (e) and (f) Feature maps after undergoing the PG module. (g) Differential mappings. (h) Label.

evaluate the efficacy of the introduced PHFA submodule, we designed aggregation of high-level features through addition combined with concatenation. Comparing the second with the fifth rows, the third with the sixth rows, and the fourth with the seventh rows demonstrates that our PHFA submodule more effectively integrates high-level features as prior information and captures positional details of changed areas to produce a semantic guidance map for localizing change regions, helping the features to better acquire precise boundary information.

2) Effectiveness of PA Module: As illustrated in Table II, when only the PA module is added, $F1$-score increased by 2.02%, 1.04%, and 2.72% on the LEVIR-CD, SYSU-CD, and WHU-CD datasets, respectively. Further ablation experiments on the two submodules of the PA module are shown in Table IV. We initially removed the embeddings of the GIDP and SAAA submodules, using FPN [11] as the decoder. When only the SAAA submodule is added, the $F1$-scores are improved by 0.52/0.47/0.54% on three CD datasets, and when only the GIDP submodule is added, the $F1$-scores are improved by 0.88/1.03/1.16%. The final model incorporates both the GIDP and SAAA modules, resulting in improvements of 1.16/1.55/1.44%, which fully validates the efficiency of the proposed PA module.

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TABLE IV

EFFECTIVENESS OF PA MODULE WHEN TRAINED ON THREE CD DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	GIDP	SAAA	LEVIR-CD	SYSU-CD	WHU-CD
(a)	×	×	90.46	82.32	93.27
(b)	×	✓	90.98	82.79	93.81
(c)	✓	×	91.34	83.35	94.43
(d)	✓	✓	**91.62**	**83.87**	**94.71**

TABLE V

COMPARISON OF DIFFERENT k VALUES. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	top-k	LEVIR-CD	SYSU-CD	WHU-CD
(a)	-	90.46	82.32	93.27
(b)	1	91.19	83.18	93.9
(c)	16	91.42	83.43	94.52
(e)	32	**91.62**	**83.87**	**94.71**
(f)	64	91.35	83.27	94.18

TABLE VI

COMPARISON OF RESULTS FOR DIFFERENT RANGES OF MOBILENETV2 BACKBONE LAYERS (i). ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	i	Params (M)	FLOPs (G)	LEVIR -CD	SYSU -CD	WHU -CD
(a)	1∼4	3.76M	10.52M	91.29	82.56	94.46
(b)	1∼5	4.13M	13.86G	91.62	**83.87**	94.71
(c)	1∼6	4.98M	17.24G	**91.83**	83.64	**94.79**

3) Discussions of Different Top-k Values: We configured different k values in the top-k operation for a comprehensive comparison to assess their contributions in CD tasks. As illustrated in Table V, when the k value is set to 1, the performance of the top-k operation is degraded to that of max pooling. Compared to this, when we increase the value to 16, there is an improvement of 0.96% on LEVIR-CD, 1.11% on SYSU-CD, and 1.25% on WHU-CD. The highest score is achieved when the k value is set to 32, resulting in improvements of 1.16% on LEVIR-CD, 1.55% on SYSU-CD, and 1.44% on WHU-CD. However, further increasing it to 64 does not yield any additional improvement in the CD results. It's worth noting that not all information is beneficial and principal for detecting changes, and in fact, irrelevant regions can have a negative impact. Therefore, our subsequent experiments are based on the best setting, which is 32.

4) Discussions of Different Ranges of MobileNetV2 Backbone Layers(i): As shown in Table VI, when the range of feature layers selected from the blocks of MobileNetV2 is 1∼4, the model's computational cost and parameter count are minimized, but the $F1$-score correspondingly decreases. When the range is 1∼5, the $F1$-score reaches its highest on the SYSU-CD dataset. When the range is 1∼6, the $F1$-score is highest on the LEVIR-CD and WHU-CD datasets, but the parameter count and computational cost are also at their maximum. Considering the tradeoff between accuracy and model size, we selected the feature layer range of 1∼5 for our subsequent experiments.

TABLE VII

EFFECTIVENESS OF PGPANET WITH DIFFERENT BACKBONES WHEN TRAINED ON THREE CD DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Backbone	Params (M)	FLOPs (G)	LEVIR -CD	SYSU -CD	WHU -CD
ResNet18	19.12M	23.56G	91.35	82.89	94.49
ResNet34	29.23M	30.22G	91.47	83.56	94.88
ResNet50	131.96M	99.36G	**91.93**	**83.98**	**95.07**

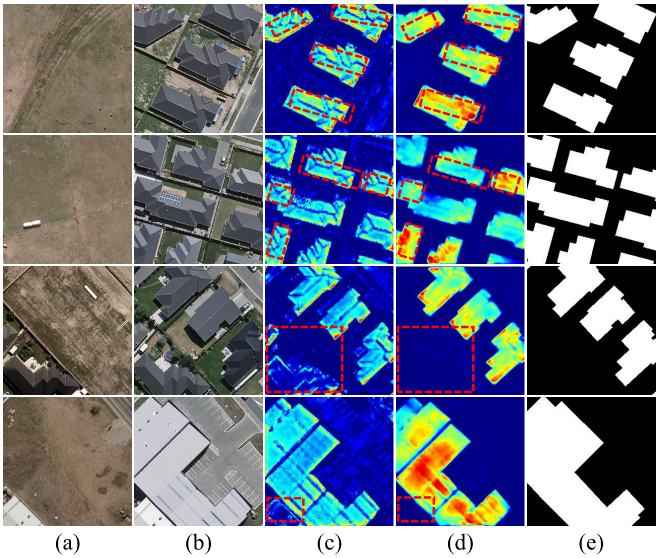


Fig. 10. Visualization of attention maps of PA module. (a) T_1 image. (b) T_2 image. (c) Input of PA module. (d) Output of the PA module. (e) Label.

5) Discussions of Different Backbones: As shown in Table VII, experimental results across three datasets indicate that PGPANet with MobileNetV2 outperforms its counterpart with ResNet18. Specifically, on the LEVIR-CD and SYSU-CD datasets, PGPANet with MobileNetV2 exhibits superior performance compared to PGPANet with ResNet34, albeit slightly trailing behind on the WHU-CD dataset. While employing ResNet50 as the backbone network results in slightly better performance, it also comes with increased parameters and computational overhead. Therefore, leveraging MobileNetV2 as the backbone network effectively balances performance and computational costs, thus validating the superiority of PGPANet.

E. Network Visualization

To more clearly illustrate the influence of our modules, we visualized heatmaps for these two modules. As depicted in the third and fourth columns of Fig. 9, the feature maps derived from the backbone network within the encoder stages typically exhibit substantial noise, which confounds the critical information of the change regions. In the fifth, sixth, and seventh columns, the outputs of our PG module eliminate background noise, enabling the acquisition of precise boundary information and compact internal representations. As shown in the first and second rows in Fig. 10, the PA module bridges the semantic gap that exists in the encoder–decoder

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structure, thereby fine structures such as roof ridges are also well identified. As shown in the third and fourth rows, our method effectively identifies pseudo-changes within changed objects under different spectra, which showcases the efficacy of the introduced modeling approaches.

V. Conclusion

In this article, we introduce a PGPANet for CD tasks in RS images. Specifically, the PG module addresses the limitations of feature extraction in traditional backbone networks by reducing noise in the encoder layers, effectively achieving accurate detection of change boundaries. The PA module resolves the issue of insufficient global information caused by traditional upsampling methods commonly employed in the decoder part of most methods, achieving excellent performance in detecting pseudo-changes with a small computational load. Furthermore, PA mitigates the semantic gap between the encoder and decoder, leading to improved extraction of fine-grained features. Experimental results from three CD datasets demonstrate that the proposed PGPANet outperforms state-of-the-art methods in terms of detection accuracy and efficiency.

It is noteworthy that although our PGPANet has achieved excellent performance with relatively low parameter and computational costs, there is still room for improvement in designing more efficient and simplified models to effectively address the challenge of deploying RS image CD on low-resource devices. Therefore, in future work, we will delve deeper into model pruning techniques. In addition, due to the high cost of acquiring labeled data, exploring semi-supervised and weakly-supervised CD methods is also worth investigating.

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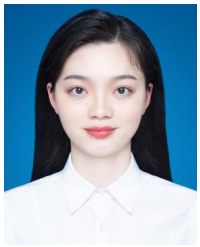
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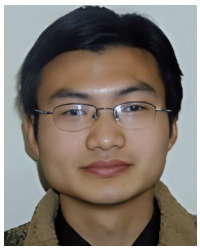
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